

Your
Team
Name

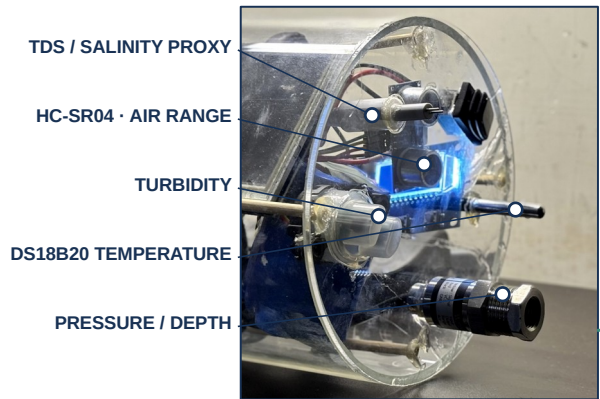
AquaSDR • TECHNICAL APPROACH

Adaptive Software-Defined Sonar Transmitter for AUVs



SENSE

Real environment • real sensors



ADC • 1-Wire • I²C

Also on board BMP390 • IMU + magnetometer (I²C) • INA219 ×2 mounted, not commissioned

ONE POLICY • TWO INPUTS



DECIDE • SYNTHESISE • STREAM

STM32F446RE embedded control • methodology & process

THE NEXT PING IS COMPUTED ON THE MCU

STM32 NUCLEO-F446RE

Sensor interface boards

MCU ADC CAPTURE
LFM example • on-board
Centre **42 kHz**
Sweep **4 kHz**
Pulse **2 ms** Hann
330/330 cmds ACKed
0 DMA underruns

- 01 ACQUIRE**
ADC • 1-Wire • I²C • stale / out-of-range inputs rejected
- 02 DECIDE**
smoothing filter + hysteresis → CLEAR / TRANSITION / MURKY
- 03 SYNTHESISE**
LFM • geometric sweep • Barker-13 × Hann / Hamming / Blackman
- 04 STREAM**
TIM6 + DMA → 12-bit DAC1 / PA4 • 1 MS/s • 0 underruns
- 06 SELF-VALIDATE**
ADC + DMA capture of the DAC output → FFT • spectrogram • matched filter

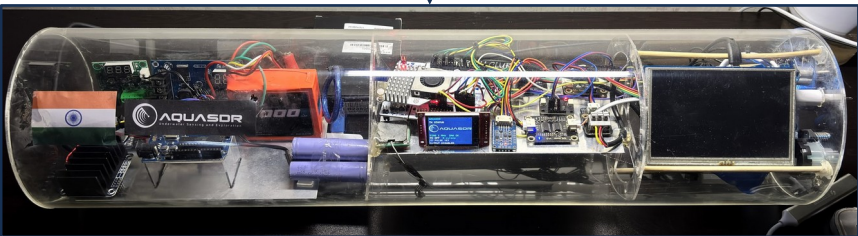
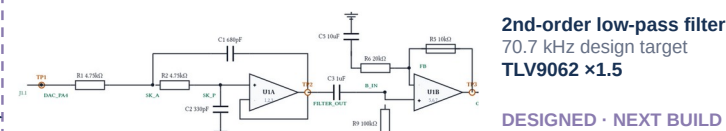
NEXT PING

Adaptive waveform parameters

	CLEAR	TRANSITION	MURKY
CENTRE	42 kHz	40 kHz	38 kHz
BANDWIDTH	4 kHz	2 kHz	1 kHz
PULSE	2 ms	4 ms	8 ms
AMPLITUDE	35 %	50 %	62 %

Water condition changes → next-ping parameters change

05 CONDITION • ANALOG FRONT END



AQUASDR PAYLOAD • PHYSICAL PROTOTYPE acrylic payload hull

External scope validation → PENDING • AUV hull-slot pod → NEXT BUILD

TECHNOLOGIES TO BE USED

Programming C / C++ • Python • JavaScript

Analog Sallen-Key LPF • TLV9062 op-amp

Embedded STM32F446RE • ADC • TIM6 • DMA • 12-bit DAC

Software React • Node.js / Express • WebSocket • OpenSCAD

Signal processing LFM • geometric sweep • Barker-13 • FFT • windowing

Hardware Turbidity • TDS • pressure • DS18B20 • IMU • BMP390 • INA219



— built / running - - - - - designed / next build